The Exact Asymptotic-Freeness Threshold for $\ell^\infty_\nu(\mu)$

A solution of the final-remarks problem in Talponen's paper

June 20, 2026

Abstract

Let $\nu < \mu$ be infinite cardinals and let $\ell^\infty_\nu(\mu)$ denote the Banach space of bounded scalar functions on μ whose supports have cardinality at most ν. Talponen asked for the least uncountable regular cardinal κ for which this space is (λ, κ)-asymptotically free. We first check that the supplied partial result, giving the sufficient condition $2^\nu < \kappa$, is correct. We then prove the exact answer: among uncountable regular κ with $\lambda < \kappa$, the space is (λ, κ)-asymptotically free if and only if $\nu < \kappa$. Consequently,

$$\kappa(\lambda, \nu, \mu) = \big(\max\{\lambda, \nu\}\big)^+.$$

The proof uses only Hahn–Banach and a sharp tail-space counterexample. In particular, the answer is independent of μ, and the hypothesis $\lambda < 2^\nu$ is not needed for the positive implication.

1 The problem

We work over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Following Talponen [1], let κ be an uncountable regular cardinal and $\lambda < \kappa$. A Banach space X is (λ, κ)-asymptotically free if, whenever $Z \subset X$ is a closed subspace with $\mathrm{dens}(Z) \leq \lambda$ and $(Z_\alpha)_{\alpha<\kappa}$ is a decreasing nested family of closed subspaces satisfying

$$Z^\perp = \bigcup_{\alpha<\kappa} Z_\alpha^\perp,$$

then, for every $x \in X$,

$$\|x + z\| \geq \|z\| \quad (z \in Z)$$

implies that there is $\alpha_0 < \kappa$ such that

$$\|x + z\| \geq \|z\| \quad (z \in Z_\alpha,\ \alpha_0 < \alpha < \kappa).$$

The space is κ-asymptotically free if this holds for every $\lambda < \kappa$.

For infinite cardinals $\nu < \mu$, put

$$\ell^\infty_\nu(\mu) = \{x \in \ell^\infty(\mu) : |\operatorname{supp}(x)| \leq \nu\},$$

with the supremum norm. Talponen's final remarks ask for the least $\kappa = \kappa(\lambda, \nu, \mu)$ such that this space is (λ, κ)-asymptotically free, in the range $\lambda < 2^\nu$ [1, Section 6].

2 Assessment of the supplied partial result

The supplied packet proves that $2^\nu < \kappa$ is sufficient. That result is correct. More precisely:

Proposition 1. *Let $\nu < \mu$ be infinite cardinals, let $\lambda < 2^\nu$, and let κ be uncountable regular with $2^\nu < \kappa$. Then $\ell^\infty_\nu(\mu)$ is (λ, κ)-asymptotically free.*

Verification. The density estimate in the packet is valid: if $|A| < 2^\nu$, then bounded rational-valued functions on supports of size at most ν form a dense family in $\ell^\infty_\nu(A)$ of cardinality at most

$$|[A]^{\le \nu}|\, \aleph_0^\nu \le |A|^\nu 2^\nu \le (2^\nu)^\nu = 2^\nu.$$

The projection-stabilization lemma is also valid in the form used in the packet, with $E_\alpha = \overline{PZ_\alpha}$ and $E = \overline{PZ}$. The identity $E^\perp = \bigcup_{\alpha<\kappa} E_\alpha^\perp$ uses the implicit fact that $Z \subset Z_\alpha$ for every α, which follows from $Z_\alpha^\perp \subset Z^\perp$. The quotient-sphere argument then forces $E_\beta = E$ at some stage $\beta < \kappa$. Applying this to the coordinate band generated by x and a dense subset of Z gives the claimed tail inequality. Thus the partial proof is sound, although its cardinal bound is not optimal. □

3 A support-synchronization lemma

The key point is that one does not need to stabilize the whole coordinate band generated by Z. Only the support of the single test vector x matters.

Lemma 2 (Coordinate correction)**.** *Let X be a linear subspace of $\ell^\infty(\Gamma)$, equipped with the supremum norm. Let $Z \subset X$ be a closed subspace and let $x \in X$ satisfy*

$$\|x + z\| \ge \|z\| \qquad (z \in Z).$$

For every $\gamma \in \Gamma$ there is a functional $u_\gamma \in X^*$ such that

$$\|u_\gamma\| \le 1, \qquad u_\gamma(x) = 0, \qquad u_\gamma(z) = z(\gamma) \quad (z \in Z).$$

Consequently, if δ_γ denotes coordinate evaluation and $g_\gamma = \delta_\gamma - u_\gamma$, then $g_\gamma \in Z^\perp$, and

$$g_\gamma(y) = 0 \quad \Longrightarrow \quad |y(\gamma)| \le \|x + y\|.$$

Proof. If $x = 0$, take $u_\gamma = \delta_\gamma|_X$. Suppose that $x \ne 0$. Then $x \notin Z$, since otherwise the choice $z = -x$ would give $0 \ge \|x\|$.

On the direct sum $W = Z \oplus \mathbb{F}x$, define

$$Q(z + tx) = z.$$

The hypothesis implies that Q is contractive. Indeed, for $t \ne 0$,

$$\|z + tx\| = |t|\,\|x + z/t\| \ge |t|\,\|z/t\| = \|z\|,$$

and the case $t = 0$ is immediate. Hence

$$v_\gamma(z + tx) = z(\gamma) = \delta_\gamma(Q(z + tx))$$

defines a functional on W of norm at most one. By Hahn–Banach it extends to a functional $u_\gamma \in X^*$ of norm at most one. It has the stated values on Z and on x.

Now $g_\gamma = \delta_\gamma - u_\gamma$ annihilates Z. If $g_\gamma(y) = 0$, then

$$|y(\gamma)| = |u_\gamma(y)| = |u_\gamma(x + y)| \le \|x + y\|.$$

□

Theorem 3 (Positive direction). *Let κ be uncountable regular, and let $X \subset \ell^\infty(\Gamma)$ be a Banach space with the inherited supremum norm. If every $x \in X$ has $|\operatorname{supp}(x)| < \kappa$, then X is κ-asymptotically free.*

Proof. Fix $\lambda < \kappa$, a closed subspace $Z \subset X$, a decreasing family $(Z_\alpha)_{\alpha<\kappa}$ satisfying $Z^\perp = \bigcup_{\alpha<\kappa} Z_\alpha^\perp$, and $x \in X$ such that $\|x + z\| \geq \|z\|$ for all $z \in Z$. Put $S = \operatorname{supp}(x)$, so $|S| < \kappa$.

For each $\gamma \in S$, apply Lemma 2 and put $g_\gamma = \delta_\gamma - u_\gamma \in Z^\perp$. Choose $\alpha_\gamma < \kappa$ with $g_\gamma \in Z_{\alpha_\gamma}^\perp$. Regularity of κ and $|S| < \kappa$ yield

$$\alpha_0 = \sup_{\gamma \in S}(\alpha_\gamma + 1) < \kappa.$$

Let $\alpha \geq \alpha_0$ and $y \in Z_\alpha$. For $\gamma \in S$, nestedness gives $g_\gamma(y) = 0$, and hence

$$|y(\gamma)| \leq \|x + y\|.$$

For $\gamma \notin S$, one has $x(\gamma) = 0$, so

$$|y(\gamma)| = |(x + y)(\gamma)| \leq \|x + y\|.$$

Taking the supremum over Γ gives $\|y\| \leq \|x + y\|$. This is the required tail inequality. Since $\lambda < \kappa$ was arbitrary, X is κ-asymptotically free. $\square$

Corollary 4. *If $\nu < \kappa$ and κ is uncountable regular, then $\ell_\nu^\infty(\mu)$ is κ-asymptotically free, for every cardinal μ.*

Proof. Every vector has support of cardinality at most $\nu < \kappa$, so Theorem 3 applies. $\square$

4 The sharp counterexample

Theorem 5 (Negative direction). *Let $\nu < \mu$ be infinite cardinals. Let κ be an uncountable regular cardinal with $\kappa \leq \nu$, and let $\lambda < \kappa$ be an infinite cardinal. Then $\ell_\nu^\infty(\mu)$ is not (λ, κ)-asymptotically free.*

Proof. Choose a set $\Gamma \subset \mu$ of cardinality κ, identify it with the ordinal κ, and regard

$$Y = c_0(\kappa)$$

as a closed subspace of $\ell_\nu^\infty(\mu)$, by extending functions by zero outside Γ. This inclusion is legitimate because every element of $c_0(\kappa)$ has countable support, hence support of size at most ν.

Set $Z = \{0\}$, and for $\alpha < \kappa$ put

$$Z_\alpha = c_0([\alpha, \kappa)) = \overline{\operatorname{span}}\{e_\xi : \alpha \leq \xi < \kappa\}.$$

These are decreasing closed subspaces. We claim that

$$X^* = Z^\perp = \bigcup_{\alpha<\kappa} Z_\alpha^\perp, \qquad X = \ell_\nu^\infty(\mu).$$

Indeed, for $f \in X^*$, the restriction $f|_Y$ belongs to $Y^* = \ell^1(\kappa)$, so its support is countable. Since κ is uncountable regular, that support is bounded in κ. Hence $f|_{Z_\alpha} = 0$ for some $\alpha < \kappa$, which proves the claim.

Finally, define $x \in X$ by

$$x(\xi) = -\frac{1}{2} \quad (\xi < \kappa), \qquad x(\eta) = 0 \quad (\eta \in \mu \setminus \Gamma).$$

Its support has cardinality $\kappa \le \nu$. The hypothesis on $Z = \{0\}$ is vacuous. But for every $\alpha < \kappa$, choosing any $\xi \in [\alpha, \kappa)$ gives $e_\xi \in Z_\alpha$ and

$$\|e_\xi\| = 1, \qquad \|x + e_\xi\| = \frac{1}{2}.$$

Thus the required inequality fails at every stage of the family, and the space is not (λ, κ)-asymptotically free. ☐

5 Exact value of the invariant

Theorem 6 (Exact answer)**.** *Let $\nu < \mu$ and λ be infinite cardinals. For every uncountable regular cardinal κ with $\lambda < \kappa$,*

$$\ell^\infty_\nu(\mu) \text{ is } (\lambda, \kappa)\text{-asymptotically free} \quad \Longleftrightarrow \quad \nu < \kappa.$$

Consequently, in the notation of Talponen's final-remarks problem,

$$\boxed{\kappa(\lambda, \nu, \mu) = \big(\max\{\lambda, \nu\}\big)^+.}$$

In particular, this holds throughout the originally requested range $\lambda < 2^\nu$.

Proof. The forward implication is Theorem 5, and the reverse implication is Corollary 4. The least uncountable regular cardinal larger than both λ and ν is the successor cardinal $\big(\max\{\lambda, \nu\}\big)^+$. ☐

Remark 7. The positive proof uses neither $\mathrm{dens}(Z) \le \lambda$ nor $\lambda < 2^\nu$, beyond the formal requirement $\lambda < \kappa$ in the definition. It also shows that, for every uncountable regular κ, the Banach space $\ell^\infty_{<\kappa}(\mu)$ is κ-asymptotically free whenever it is complete. Thus the same argument strengthens the strongly-inaccessible hypothesis appearing in Corollary 4.5 of [1].

6 Literature check

A targeted search was made for the exact paper title, arXiv:1604.04408, the phrases "asymptotically free Banach space" and "(λ, κ)-asymptotically free", and the specific notation $\ell^\infty_\nu(\mu)$, together with citation-oriented searches for Talponen's paper. The searches located the source paper and indexing or announcement pages, but no later paper giving either the exact invariant above or a counterexample resolving the final-remarks question. This is a targeted rather than logically exhaustive bibliographic search, so the claim here is only that no prior resolution was found.

References

[1] J. Talponen, *Unconditional and bimonotone structures in high density Banach spaces*, arXiv:1604.04408v2, 2016.